

Curriculum Vitae

Personal Data

Pietro Benedusi, Ph.D.

Research fellow at Euler Institute, Università della Svizzera Italiana (USI), Lugano, Switzerland

pietro.benedusi@usi.ch

<https://pietrobe.github.io/>

Research Interests

Numerical analysis, scientific computing, multigrid and time-parallel methods, high-performance computing, large-scale applications in radiative transfer and computational electrophysiology, scientific machine learning.

Research Positions

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| 4/2025 – present | Scientific collaborator (100%) at the “Istituto ricerche solari Aldo e Cele Daccò” (IRSOL) and lecturer at USI, Lugano. |
| 6/2020 – 3/2025 | Postdoctoral researcher at the Euler Institute, USI, Lugano. |
| 4/2022 – 4/2024 | Postdoctoral researcher at the Simula Research Laboratory as a computational neuroscientist and HPC expert, Oslo. |
| 8/2019 – 10/2019 | Research visit at Berkeley Lab. |
| 1/2014 – 9/2015 | Student assistant at USI, developing scalable algorithms for astrophysical simulations. |

Education

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| 10/2015 – 5/2020 | PhD in Computational Science, USI, under the supervision of Prof. Rolf Krause. Research on numerical schemes for massively parallel PDE solvers exploiting time parallelism. |
| 9/2013 – 9/2015 | Master’s degree with summa cum laude in Applied Mathematics and Computational Science at USI with thesis “A Parallel Multigrid Solver for Time-periodic Incompressible Navier-Stokes Equations” joint with ETH Zurich. |
| 9/2009 – 4/2013 | Bachelor’s degree in Physics at University of Pavia with thesis: “Neutrino Oscillations”. |

Selected Achievements and Activities

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| 3/11/2025 | Italian habilitation as Associate Professor in Numerical Analysis (unanimity of the commission), ASN 23. |
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2025	PASC26 conference, Papers Program Committee Member (Computational Methods and Applied Mathematics).
2025	Offered a tenure track professorship at UM6P, Morocco.
2023	Co-organizing mini-symposium “Advances in Iterative Solution Methods for Solving Coupled Problems” at COUPLED PROBLEMS 2025, Villasimius, Italy.
2025	Granted, as Co-PI, 1 Million CPU hours for the project “Towards extreme-scale simulations of polarized radiative transfer for solar physics” at the Barcelona Supercomputing Center.
2024	Granted, as Co-PI, 2 Million CPU hours for the project “Modeling the polarization of the chromospheric CaI 4227 Å line in 3D models of the solar atmosphere with partial frequency redistribution” at the Barcelona Supercomputing Center.
2023	Organizing mini-symposium “High-performance computing tools for mixed dimensional problems ” at SIAM CSE23, Amsterdam, Netherlands.
2022	Organizing mini-symposium “Scalable multilevel and multi-fidelity strategies for radiative transfer problems” at the International Multigrid Conference, Lugano, Switzerland.
2022	Granted a development proposal for computing resources in the Swiss super computing centre, in the context of solar simulations.
2019	Winner of the SIAM Travel Award grant to attend the CSE19 conference held in Spokane, USA.
2017	Organization of the summer school at USI: “Generalized locally Toeplitz sequences: a spectral analysis tool for discretized differential equations”, Lugano, Switzerland.
2015	Recipient of the “USIimpresa” award and grant for best student in the Faculty of Informatics.

Teaching and supervision

2025	Lecturer of the master course “Scientific Machine Learning” at USI (graded 9.4/10 by students).
2025	PhD course “Advanced Tools in Computational Physiology ” at Insubria University, 16h.
2022 – present	Co-supervision of Halvor Herlyng, PhD candidate in the department of Scientific Computing at Simula.
2023 – present	Co-supervising four Master’s theses in Computational Science (USI and Simula)

2024	Lecturer at the doctoral Summer School in Computational Physiology at the Simula Research Laboratory and for the MICROCARD workshop. Title: “Cell-by-cell modeling tutorial”, 3 hours.
2023	Lecturer at the doctoral Summer School in Computational Physiology at the Simula Research Laboratory (joint event between the University of California San Diego and the University of Oslo). Title “The EMI Model: applications in cardiac tissue and neuroscience and code-based exercises”, 5 hours.
2020 – 2023	Co-supervision of Simone Riva, PhD in Computational Science at USI in August 2023.
2017 – 2020	Teaching assistant for the following courses: Functional and Numerical Analysis, Theory of Computation, Fast Solvers, Introduction to PDEs, Linear Algebra, and Introduction to Computational Science at USI, ~ 90 hours each.

Conference Presentations and Invited Talks

4/2026	Swiss Numerics Day, Zurich, Switzerland.
9/2025	<u>Invited</u> speaker at the 11th Solar Polarization Workshop, Prague, Czech Republic.
5/2025	Speaker at the XI International Conference on Coupled Problems in Science and Engineering, Villasimius, Italy.
2/2025	Speaker at the International Multigrid Conference, KAUST, Saudi Arabia.
2/2025	Speaker at the “Due Giorni di Algebra Lineare Numerica” conference, Pisa, Italy.
12/2024	<u>Invited</u> speaker at the Math.&Decisions conference, Rabat, Morocco.
6/2024	Speaker at the FEniCs conference, Oslo, Norway.
6/2024	Speaker at ECCOMAS conference, Lisbon, Portugal.
1/2024	Speaker at DD28, KAUST, Saudi Arabia.
11/2023	Speaker at SfN, Washington, USA.
8/2023	Speaker at ICIAM, Tokyo, Japan.
7/2023	Speaker at the MICROCARD workshop, Strasbourg, France.
11/2022	Speaker at SPW10, Kyoto, Japan.
9/2022	Speaker at GIMC SIMAI YOUNG 2022, Pavia, Italy.
8/2022	Speaker at IMG2022, Lugano, Switzerland.

7/2022	Speaker at the 11th Parallel-in-Time Workshop, CIRM, France.
6/2022	Speaker at the ECCOMAS conference in Oslo, Norway.
2/2022	Speaker at the SIAM PP21 conference, held online.
9/2021	Speaker at the SIMAI 2021 conference in Parma, Italy.
8/2021	Speaker at the USNCCM16 conference and at the 10th Parallel-in-Time Workshop, both held online.
3/2021	Speaker at the SIAM CSE conference, held online.
12/2020	Speaker at the FOMICS-DADSI winter school on cardiac simulations, held online.
6/2020	Speaker at the 9th Parallel-in-Time Workshop, held online.
5/2019	Speaker at the 8th Parallel-in-Time Workshop in Bielefeld, Germany.
2/2019	Speaker at CSE19 conference in Spokane, USA.
7/2018	Speaker at DD25 conference in St John's, Canada.
4/2018	Poster at the Swiss Numerics Day, ETH, Zurich, Switzerland.
10/2017	Speaker at the 6th Parallel-in-Time Workshop, Ascona, Switzerland.
6/2017	Speaker at LSSC17 conference in Sozopol, Bulgaria.
3/2017	Speaker at the 18th Copper Mountain Conference On Multigrid Methods, Colorado.
2/2017	Speaker at DD24 conference in Svalbard, Norway.
6/2016	Speaker at MAFELAP conference, London.
6/2016	Speaker at ECCOMAS conference, Crete.
3/2016	Speaker at SIAM PP16 conference, Paris.
12/2015	Speaker at the CBC Workshop on Dynamic Adaptivity, Simula, Oslo.

Computational Tools

C/C++, MATLAB, Python. Good knowledge of protocols for parallel computing (MPI and OpenMP) and HPC frameworks (e.g. Slurm). Experience with finite element libraries (LibMesh and FEniCS), meshing tools, numerical packages (PETSc, Trilinos), and PyTorch.

Publications

- [1] P. Benedusi, P. Ferrari, M. Causemann, and S. Serra-Capizzano, "Dense cell-by-cell systems of PDEs: approximation, spectral analysis, and preconditioning," *BIT Numerical Mathematics*, vol. 66, no. 53, 2026.

- [2] M. Tonarelli, S. Riva, P. Benedusi, F. Ferrandi, and R. Krause, “Adaptive multidimensional quadrature on multi-gpu systems,” in *Proceedings of the 28th International Conference on Domain Decomposition Methods (DD29)*, 2026.
- [3] F. Riva, G. Janett, L. Belluzzi, T. del Pino Alemán, E. A. Ballester, J. T. Bueno, P. Benedusi, S. Riva, and R. Krause, “A numerical approach for modelling the polarisation signals of strong resonance lines with partial frequency redistribution-applications to two-term atoms and plane-parallel atmospheres,” *Astronomy & Astrophysics*, vol. 699, p. A233, 2025.
- [4] A. J. Ellingsrud, P. Benedusi, and M. Kuchta, “A splitting, discontinuous Galerkin solver for the cell-by-cell electroneutral Nernst–Planck framework,” *SIAM Journal on Scientific Computing*, vol. 47, no. 2, pp. B477–B504, 2025.
- [5] A. Gatti, P. Benedusi, and S. Pezzuto, “Cell-by-cell modelling of electrical signaling in myelinated axons,” in *Proceedings of the 28th International Conference on Domain Decomposition Methods (DD28)*, 2024.
- [6] P. Benedusi, A. J. Ellingsrud, H. Herlyng, and M. E. Rognes, “Scalable approximation and solvers for ionic electrodiffusion in cellular geometries,” *SIAM Journal on Scientific Computing*, vol. 46, no. 5, pp. B725–B751, 2024.
- [7] L. Belluzzi, S. Riva, G. Janett, N. Guerreiro, F. Riva, P. Benedusi, T. del Pino Alemán, E. A. Ballester, J. T. Bueno, and J. Štěpán, “Accurate PRD modeling of the forward-scattering Hanle effect in the chromospheric Ca I 4227 Å line,” *Astronomy & Astrophysics*, vol. 691, p. A278, 2024.
- [8] G. Janett, P. Benedusi, and F. Riva, “Numerical solutions to linear transfer problems of polarized radiation-IV. efficient preconditioning in a physics-based framework,” *Astronomy & Astrophysics*, vol. 682, p. A68, 2024.
- [9] P. Benedusi, P. Ferrari, M. Rognes, and S. Serra-Capizzano, “Modeling excitable cells with the EMI equations: spectral analysis and iterative solution strategy,” *Journal of Scientific Computing*, vol. 98, no. 3, p. 58, 2024.
- [10] N. Guerreiro, G. Janett, S. Riva, P. Benedusi, and L. Belluzzi, “Modeling the scattering polarization in the solar Ca I 4227Å line with angle-dependent PRD effects and bulk velocities,” *Astronomy & Astrophysics*, vol. 683, p. A207, 2024.
- [11] S. Riva, N. Guerreiro, G. Janett, D. Rossinelli, P. Benedusi, R. Krause, and L. Belluzzi, “Assessment of the CRD approximation for the observer’s frame RIII redistribution matrix,” *Astronomy & Astrophysics*, vol. 679, p. A87, 2023.
- [12] P. Benedusi, S. Riva, P. Zulian, J. Štěpán, L. Belluzzi, and R. Krause, “Scalable matrix-free solver for 3D transfer of polarized radiation in stellar atmospheres,” *Journal of Computational Physics*, 2023.
- [13] P. Benedusi, G. Janett, S. Riva, L. Belluzzi, and R. Krause, “Numerical solutions to linear transfer problems of polarized radiation-III. Parallel preconditioned

- Krylov solver tailored for modeling PRD effects,” *Astronomy & Astrophysics*, 2022.
- [14] P. Benedusi, G. Janett, L. Belluzzi, and R. Krause, “Numerical solutions to linear transfer problems of polarized radiation-II. Krylov methods and matrix-free implementation,” *Astronomy & Astrophysics*, vol. 655, p. A88, 2021.
 - [15] G. Janett, P. Benedusi, L. Belluzzi, and R. Krause, “Numerical solutions to linear transfer problems of polarized radiation-I. Algebraic formulation and stationary iterative methods,” *Astronomy & Astrophysics*, vol. 655, p. A87, 2021.
 - [16] P. Benedusi, M. L. Minion, and R. Krause, “An experimental comparison of a space-time multigrid method with pfasst for a reaction-diffusion problem,” *Computers & Mathematics with Applications*, vol. 99, pp. 162–170, 2021.
 - [17] P. Benedusi, P. Ferrari, C. Garoni, R. Krause, and S. Serra-Capizzano, “Fast parallel solver for the space-time IgA-DG discretization of the diffusion equation,” *Journal of Scientific Computing*, vol. 89, no. 1, pp. 1–21, 2021.
 - [18] S. B. Bader, P. Benedusi, A. Quaglino, P. Zulian, and R. Krause, “Space-time multilevel Monte Carlo methods and their application to cardiac electrophysiology,” *Journal of Computational Physics*, vol. 433, p. 110164, 2021.
 - [19] P. Benedusi, C. Garoni, R. Krause, X. Li, and S. Serra-Capizzano, “Space-time FE-DG discretization of the anisotropic diffusion equation in any dimension: the spectral symbol,” *SIAM Journal on Matrix Analysis and Applications*, vol. 39, no. 3, pp. 1383–1420, 2018.
 - [20] A. Kreienbuehl, P. Benedusi, D. Ruprecht, and R. Krause, “Time-parallel Gravitational Collapse Simulation,” *Communications in Applied Mathematics and Computational Science*, vol. 12, no. 1, pp. 109–128, 2017.
 - [21] L. Grasedyck, C. Löbbert, G. Wittum, A. Nägele, V. Schulz, M. Siebenborn, R. Krause, P. Benedusi, U. Küster, and B. Dick, “Space and time parallel multigrid for optimization and uncertainty quantification in PDE simulations,” in *Software for Exascale Computing-SPPEXA 2013-2015*, pp. 507–523, Springer, 2016.
 - [22] P. Benedusi, D. Hupp, P. Arbenz, and R. Krause, “A parallel multigrid solver for time-periodic incompressible Navier–Stokes equations in 3D,” in *Numerical Mathematics and Advanced Applications ENUMATH 2015*, pp. 265–273, Springer, 2016.
 - [23] P. Benedusi, *Parallel space-time multilevel methods with application to electrophysiology*. PhD thesis, Università della Svizzera italiana, 2020.

Last updated: July 2026